PAVAN KUMAR MATHI

Embedded Systems & Edge AI Engineer | Firmware · Computer Vision · LLM Agents · Full-Stack

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SUMMARY

Final-year B.Tech ECE student (2027) who builds complete systems, from Embedded C firmware to the AI and web software on top. Building A.I.R, an offline AI rover on the Arduino UNO Q (YOLOv8 mAP@50 0.951); built KAISER, a local AI agent with 41 tools and 365 tests; wrote firmware for four ARM7 LPC2148 systems; and shipped a live e-commerce site. Seeking an entry-level embedded, edge-AI or software role.

TECHNICAL SKILLS

Embedded & Hardware: Embedded C, ARM7 LPC2148, Arduino UNO / UNO Q, ESP32, UART / I2C / SPI / ADC, timers, interrupts, RTC, watchdog, PCB design, Keil

AI / ML: YOLOv8, OpenCV, MediaPipe, PyTorch, Transformers, LLM agents, RAG, Ollama, LoRA, Whisper / Piper

Software & Web: Python, C, JavaScript, SQL, Flask, SQLAlchemy, PostgreSQL, REST APIs, Razorpay

Tools: Git, GitHub, Vercel, Linux (Debian), Fusion 360, AutoCAD, Google Colab, Blynk, n8n

PROJECTS

A.I.R – Artificial Intelligence Rover (Final-Year Project) | *Arduino UNO Q, Embedded C, Python, YOLOv8, Ollama*

- Offline companion rover: MCU firmware (motor control, sensors, emergency stop) plus a Linux side running 8 on-device AI models, local LLM chat and speech in/out.
- Trained a YOLOv8n detector on 592 images (mAP@50 0.951, recall 0.948); tested the AI-to-MCU control loop in a simulator, reaching 0 wall collisions in a 5-minute run.

KAISER – Offline Personal AI Agent | *Python, Ollama, RAG, Whisper*

- Voice-activated agent that plans, executes and verifies tasks with 41 tools on local LLMs: self-repairing coding loop, cited web research and PDF Q&A.
- Sandboxed execution, SSRF-safe browser, HMAC audit log and signed phone commands; 365 automated tests passing.

ARM7 LPC2148 Firmware – Countdown Timer, Secure Access, Smart Security, Multi-Peripheral | *Embedded C, Keil*

- Four register-level Embedded C systems using RTC, timers, interrupts, ADC, UART, I2C EEPROM, SPI and watchdog; fixed random resets caused by spurious IRQs; tested on the lab board.

Multimodal Disaster Response Decision Support | *PyTorch, DistilBERT, ResNet50, PuLP*

- Text + image classifiers, two-axis severity scoring and fair LP resource allocation with LLM explanations: +24.8% priority-weighted coverage, under 100 ms per report on CPU.

Full-Stack Web Apps – Ladies Fancy Store (live) and Back Benchers Food Stall | *Flask, PostgreSQL, Razorpay, Vercel*

- Built and deployed a production e-commerce site for a local shop (catalogue, cart, Razorpay + COD, admin panel) and a food-stall ordering system with verified payment webhooks.

ESP32-CAM Surveillance Robot | *ESP32-CAM, Arduino, Motor Drivers, Wi-Fi*

- Wi-Fi robot car with MJPEG live video, a browser control dashboard and PWM speed control; adding on-board human detection with snapshot alerts.

INTERNSHIPS

Embedded Systems & IoT Intern – Sreshta Semicon Technologies Pvt. Ltd. **Jun 2026 – Jul 2026**

- Hands-on training and project work in embedded development and IoT architecture, connecting sensor data from hardware to software systems.

PCB Design & Electronics Intern – Pantech Solutions **May 2026 – Jun 2026**

- PCB layout workflow, circuit design practice and 8255-based microcontroller board fundamentals.

EDUCATION

B.Tech, Electronics and Communication Engineering – MVGR College of Engineering (A), Vizianagaram **2023 – 2027**

Intermediate (MPC) – Sri Chaitanya Junior College, Vizianagaram **2021 – 2023**

ACHIEVEMENTS & LEADERSHIP

- Team Lead – ranked 28th of 100+ teams, Autodesk 36-hour hackathon (ANITS, Jan 2026); built the prototype's Fusion 360 CAD.
- SPOC Coordinator – organised an electronics workshop at MVGR's Aadhrita tech fest and mentored 50+ participants.

Languages: English, Telugu, Hindi